

The t -percentile for $a = \alpha/2 = .025$ and $df = 20 + 19 - 2 = 37$ is not listed in Table 2 of the Appendix, but taking the labeled value for the nearest df less than 37 ($df = 35$), we have $t_{.025} = 2.030$. A 95% confidence interval for the difference in mean incomes for the two districts is of the form

$$\bar{y}_1 - \bar{y}_2 \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Substituting into the formula, we obtain

$$1.49 \pm 2.030(2.77) \sqrt{\frac{1}{20} + \frac{1}{19}} \quad \text{or} \quad 1.49 \pm 1.80$$

Thus, we estimate the difference in mean incomes to lie somewhere in the interval from $-.31$ to 3.29 . If we multiply these limits by \$1,000, the confidence interval for the difference in mean incomes is $-\$310$ to $\$3,290$. This interval includes both positive and negative values for $\mu_1 - \mu_2$, so we are unable to determine whether the mean income for District 1 is larger or smaller than the mean income for District 2.

We can also test a hypothesis about the difference between two population means. As with any test procedure, we begin by specifying a research hypothesis for the difference in population means. Thus, we might, for example, specify that the difference $\mu_1 - \mu_2$ is greater than some value D_0 . (Note: D_0 will often be 0.) The entire test procedure is summarized here.

A Statistical Test for $\mu_1 - \mu_2$, Independent Samples

The assumptions under which the test will be valid are the same as were required for constructing the confidence interval on $\mu_1 - \mu_2$: population distributions are normal with equal variances and the two random samples are independent.

$$H_0: \begin{array}{l} 1. \mu_1 - \mu_2 \leq D_0 \\ 2. \mu_1 - \mu_2 \geq D_0 \\ 3. \mu_1 - \mu_2 = D_0 \end{array} \quad (D_0 \text{ is a specified value, often } 0)$$

$$H_a: \begin{array}{l} 1. \mu_1 - \mu_2 > D_0 \\ 2. \mu_1 - \mu_2 < D_0 \\ 3. \mu_1 - \mu_2 \neq D_0 \end{array}$$

$$\text{T.S.: } t = \frac{(\bar{y}_1 - \bar{y}_2) - D_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- R.R.: For a level α , Type I error rate and with $df = n_1 + n_2 - 2$,
1. Reject H_0 if $t \geq t_{\alpha}$.
 2. Reject H_0 if $t \leq -t_{\alpha}$.
 3. Reject H_0 if $|t| \geq t_{\alpha/2}$.

Check assumptions and draw conclusions.